NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) — path to success

# RACE
Regular Analysis through Continuous Exercise

JEE (Main + Advanced) 2023
ENTHUSIAST + LEADER COURSE

**SCORE ADVANCED** | **MATHEMATICS** | **M.M. 77**

REVISION ASSINGMENT # 12 (CONTINUITY, DIFFERENTIABILITY & MOD) TIME : 70 MIN.

## SECTION–I

**Straight Objective Type** (3 Marks each, –1 for wrong answer)

**1.** Let g is the inverse function of differentiable function f & $f'(x)=\frac{x^{10}}{1+x^2}$. If g (2) = a then g'(2) is equal to-

(A) $\frac{5}{2^{10}}$ (B) $\frac{1+a^2}{a^{10}}$ (C) $\frac{a^{10}}{1+a^2}$ (D) $\frac{1+a^{10}}{a^2}$

**2.** If y = tanθ + cotθ, x = tanθ – cotθ, where θ is a parameter then $y^3\frac{d^2y}{dx^2}$ is equal to-

(A) 2 (B) –2 (C) 4 (D) –4

**3.** If $f(x)=\sqrt{x+3-4\sqrt{x-1}}+\sqrt{x+8-6\sqrt{(x-1)}}$, then $f'(x)$ at x = 1.5 is -

(A) 0 (B) $-\sqrt{2}$ (C) $-\sqrt{3}$ (D) –4

**4.** If the normal to the curve y = $f$ (x) at x = 0 be given by 3x – y + 3 = 0, then the value of $\lim_{x\to1}(x-1)^2\left\{f((x-1)^2)-5f(4(x-1)^2)+4f(7(x-1)^2)\right\}^{-1}$ is -

(A) $-\frac{1}{3}$ (B) $\frac{1}{3}$ (C) 3 (D) –3

**5.** In order that the function $f(x)=(x+1)^{\cot x}$ be continuous at x = 0, then f(0) must be equal to :

(A) 0 (B) $\frac{1}{e}$ (C) 1 (D) e

**6.** $\lim_{x\to1}(x-1)\ell n(\sin^{-1}(\ell nx))$ is equal to :-

(A) 0 (B) $\frac{\pi}{2}$ (C) $\ell n\frac{\pi}{2}$ (D) 1

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**7.** Let $f(x)=\begin{cases} g(x).\cos\frac{1}{x} & \text{if } x\neq 0 \\ 0 & \text{if } x=0 \end{cases}$ where g(x) is an even function differentiable at x = 0, passing through the origin. Then f'(0)

(A) is equal to 1 (B) is equal to 0 (C) is equal to 2 (D) does not exist

**8.** $\lim\limits_{x\to\infty}\dfrac{\cos^{-1}\left(\sqrt{x+\sqrt{x+\sqrt{x}}}-\sqrt{x+\sqrt{x}}\right)}{\sec^{-1}\left(-x^2\ell n\left(\frac{x^2+2}{x^2+1}\right)\right)}$ is equal to -

(A) 1 (B) $\frac{\pi}{2}$ (C) $\frac{1}{2}$ (D) $\pi$

**9.** Let $g:R\to R$ be a differentiable function such that, $\lim\limits_{\alpha\to\frac{\pi}{2}} g(2009+\cos\alpha)=3$

and $\lim\limits_{n\to\infty} n\left(g\left(2009+\frac{1}{n}\right)-g(2009)\right)=27$, then $\lim\limits_{t\to 0^+}\left(\frac{g(2009)}{g(2009+t)}\right)^{1/t}$ is :-

(A) $e^3$ (B) $e^2$ (C) $e^{-2}$ (D) $e^{-9}$

**10.** Let $f(u)=\frac{1}{\ell n|u|-1}$, $u=\frac{x+e}{x+1}$. Number of values of x in domain for which $f(u)$ is discontinous is -

(A) 2 (B) 1 (C) 3 (D) 0

**Multiple Correct Answer Type** (4 Marks each, –1 for wrong answer)

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**11.** If g is inverse of $f$ and $f(x)=x^2+3x-3$ $(x>0)$ then g'(1) equals -

(A) $\frac{1}{2g(1)+3}$ (B) –1 (C) $\frac{1}{5}$ (D) $\frac{-f'(1)}{(f(1))^2}$

**12.** If $\lim\limits_{x\to\infty}\left(\frac{ax+2}{\sqrt{4+x^2}}\right)^{\frac{ax^3+1}{x^2+1}}=L$, where $a\geq 0$ & L is finite number, then -

(A) a = 0, L = 1 (B) a = 1, $L=e^2$ (C) $0<a<1$, L = 0 (D) $0\leq a<1$, L = 0

**13.** $f(x)=(\sin^{-1}x)^2.\cos(1/x)$ if $x\neq 0$, f(0) = 0, f(x) is :

(A) continuous no where in $-1\leq x\leq 1$ (B) continuous every where in $-1\leq x\leq 1, x\neq 0$

(C) differentiable no where in $-1\leq x\leq 1$ (D) differentiable everywhere $-1<x<1, x\neq 0$

**14.** If $f(x)=\cos\left[\frac{\pi}{x}\right]\cos\left(\frac{\pi}{2}(x-1)\right)$; where [x] is the greatest integer function of x, then f(x) is continuous at-

(A) x = 0 (B) x = 1 (C) x = 2 (D) none of these

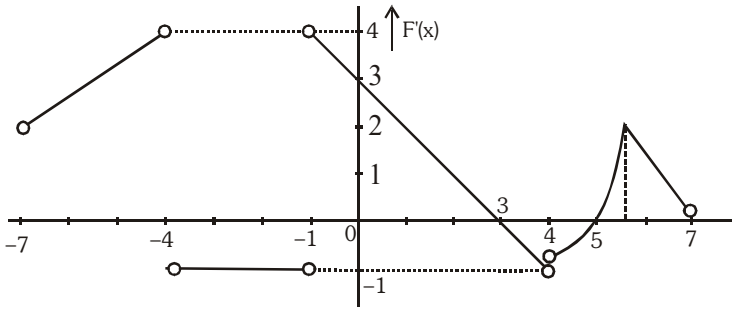
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**Linked Comprehension Type (Single Correct Answer Type)** (3 Marks each, –1 for wrong answer)

**Paragraph for Question 15 to 16**

Graph of the derivative of a continuous function F(x) in (–7, 7) is given below




On the basis of above information answer the following :

**15.** F(x) is a linear function in :-

(A) (–7, –4) (B) (–4, –1) (C) (–1, 4) (D) $\left(\frac{11}{2},7\right)$

**16.** F'(x) does not exist at :-

(A) 1 points (B) 2 points (C) 3 points (D) 4 points

**SECTION–III**

**Numerical Grid Type (Single digit Ranging from 0 to 9)** (4 Marks each, –1 for wrong answer)

**1.** If $f : R \to R$ is a function such that $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(4)$, then find $f(1)$

**2.** Let $y(x) = \sin(x - \sin x)$ and $y''(x)$ denotes the second order derivative of $y(x)$ w.r.t.x. The value of $\left[y''\left(\frac{\pi}{2}\right) + \cos 1\right]$ is- (where [ . ] denotes the greatest integer function).

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**3.** If function $f(x) = \begin{cases} \frac{\sqrt[a]{7+x^3} - \sqrt{3+x^2}}{(x-1)}, & x \neq 1 \\ b, & x = 1 \end{cases}$ (where $a \in N$ and $b \in R$) is continuous at x = 1, then value of a + 4b is

**4.** Number of points at which $f(x) = \sin\left([x]\frac{\pi}{2}\right)\text{sgn}(x^2 - 3x + 2)$ is discontinuous in $x \in [0, 3]$, is (where [.] and sgn (.) denotes greatest integer and signum function respectively)

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## SECTION–IV

### Matrix Match Type (One or More than one option correct)

For each entry in **Column-I** , +2 If only the bubble(s) corresponding to all the correct matche(es) is (are) darkened, 0 In none of the bubbles is darkened, –1 In all other cases

**1.** Consider a function $f$ (x) in [0, 2π]

defined as $f(x) = \begin{cases} [\sin x] + [\cos x], & 0 \le x \le \pi \\ [\sin x] - [\cos x], & \pi < x \le 2\pi \end{cases}$

where [.] denotes greatest integer function.

On the basis of above match the entries of column-I with the corresponding entries in column-II.

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| (A) | Number of points where $f$ (x) is discontinuous is equal to | (P) | 2 |
| (B) | Number of points, where $\lim_{x\to a} f(x)$ does not exist is equal to (where $a \in (0, 2\pi)$) | (Q) | 3 |
| (C) | Number of points where $f$ (x) is non derivable is equal to | (R) | 4 |
| (D) | Minimum value of $f$(x) is m then \|m\| is | (S) | prime |
| | | (T) | neither composite nor prime |

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**REVISION ASSINGMENT # 11** **LIMITS** **MATHEMATICS**

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| SECTION-I | Q. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| | A. | A | D | B | B | D | B | D | A,B,C | B,C,D | A,C,D |
| | Q. | 11 | 12 | | | | | | | | |
| | A. | B | A | | | | | | | | |
| SECTION-III | Q. | 1 | 2 | 3 | | | | | | | |
| | A. | 3 | 5 | 2 | | | | | | | |
| SECTION-IV | Q.1 | A | B | C | D | Q.2 | A | B | C | D | |
| | | R | P | S | Q | | Q | P | S | R | |


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